Wednesday, June 10, 2026 8:57 PM

DECISION TREES ⇒

* Agenda ⇒
  - → use case
  - → DTree intuition
  - → working DTree
  - → Entropy, info gain
  - → Sklearn code

Reg^n + Classification

1st → 2nd

* Employee ATTRITION →



* AIRTEL → JIO
  - Employees

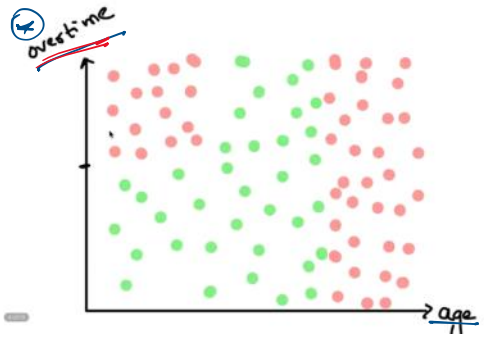
we want to predict why employees are leaving

Retain

Employee Attⁿ
- Task 1: Predict attrition or not
- Task 2: find key factors *

* Red → people left
  - green → stay

0 → Not leaving
1 → leaving



2 features =

↓

log Reg^n

↓

X

KNN

↓

YES

↳ Disadvantage

Time

↑

DTree → KNN — a — Log Reg^n

accuracy ✓ — time complexity ✓

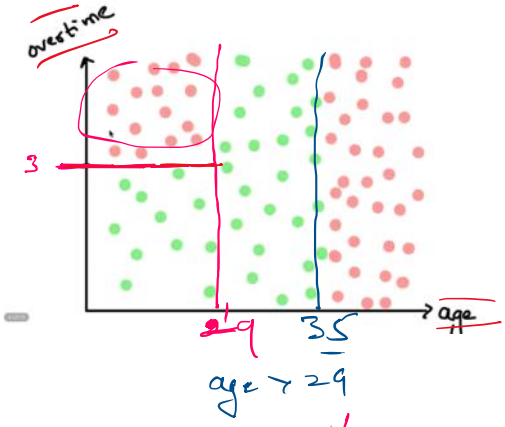
↑

Accuracy

(*) Creates Decision Boundaries

↳ axis parallel Boundaries

(*) (*)



ques →

* age < 29 or age > 29

age < 29

↓

NO when * age < 29

age > 29

if overtime > 3 hours & age < 29

↓

leaving

if

age B/w 29 & 35

↳ Pr[illegible]

age b/w < = 35

leaving

Stay

age > 35 → Leave

Decision Trees ⇒

→ Non linear model

→ Bunch of if else conditions

→ creates axis || Hyperplanes → parallel to

→ create Homogeneous Regions

⊛ similar class

⇒ d features ⇒

* Time

& optimised

.csv, .excel

.pickle

Feature

* Feature selection ⇒

Tree → Hierarchy structure

Root Node

age < 29

Branches

Yes

No

Child Nodes

internal nodes

overtime < 3

age > 33

Yes

No

Yes

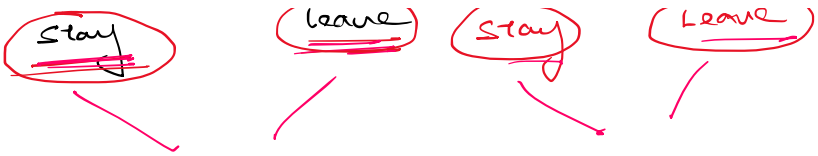
No

Stay

leave

Stay

Leave



Leaf nodes

```
if age < 29:
    if overtime < 3:
        stay
    else
        Leave
else:
    if age < 35
        stay
    else
        Leave
```

* if else code ⇒

stay → age = 29 & overtime = 2

* ⇒ Stay

Diagonal

if - else

Quiz time!

Time Left: 6s

Statement I : Topmost node is called leaf node Statement II: Topmost node is called root node. Statement III: Bottom nodes are called root nodes. Statement IV: Nodes in between root and leaf are called decision nodes/ internal nodes. Which of the following statements are true ?

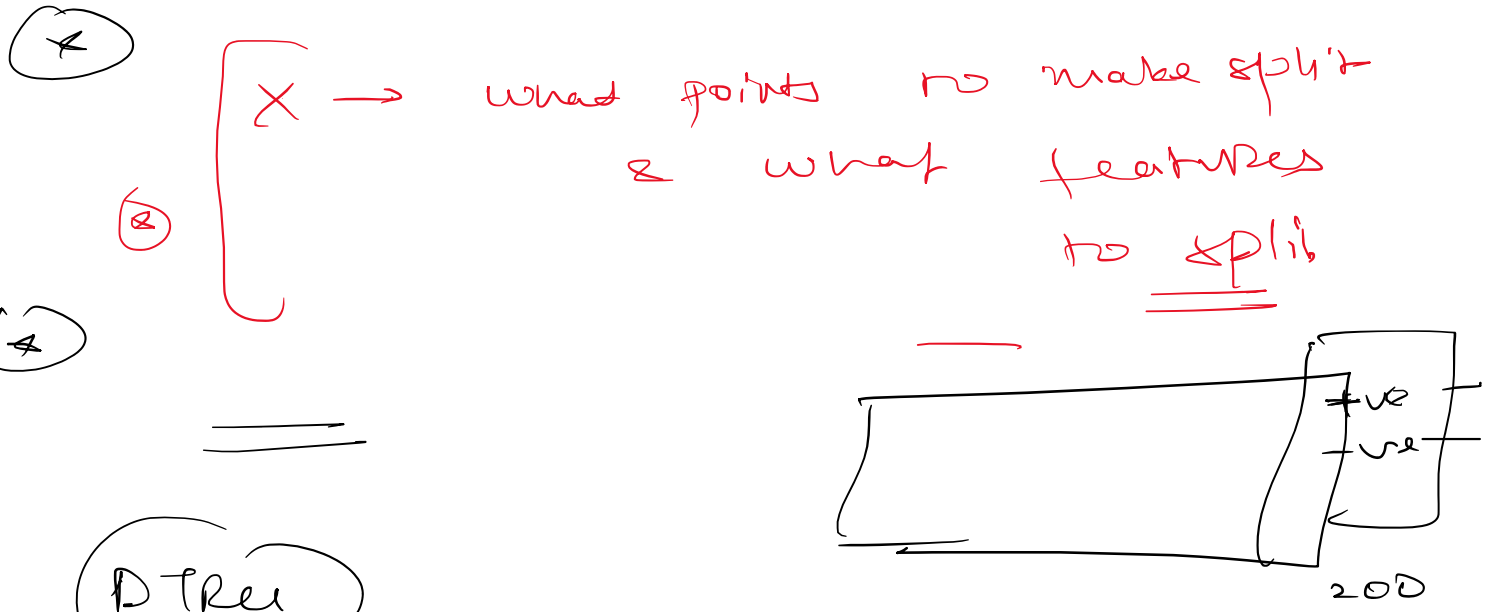
II

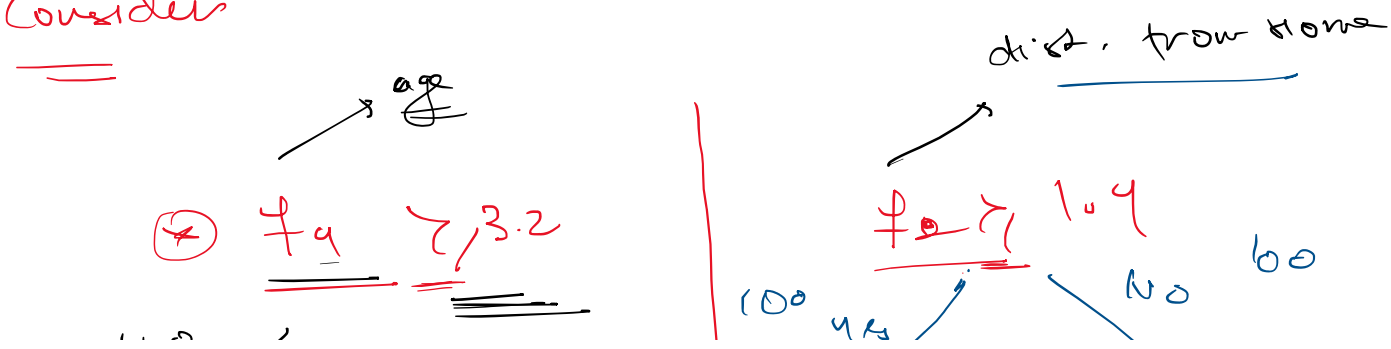
43 users have participated

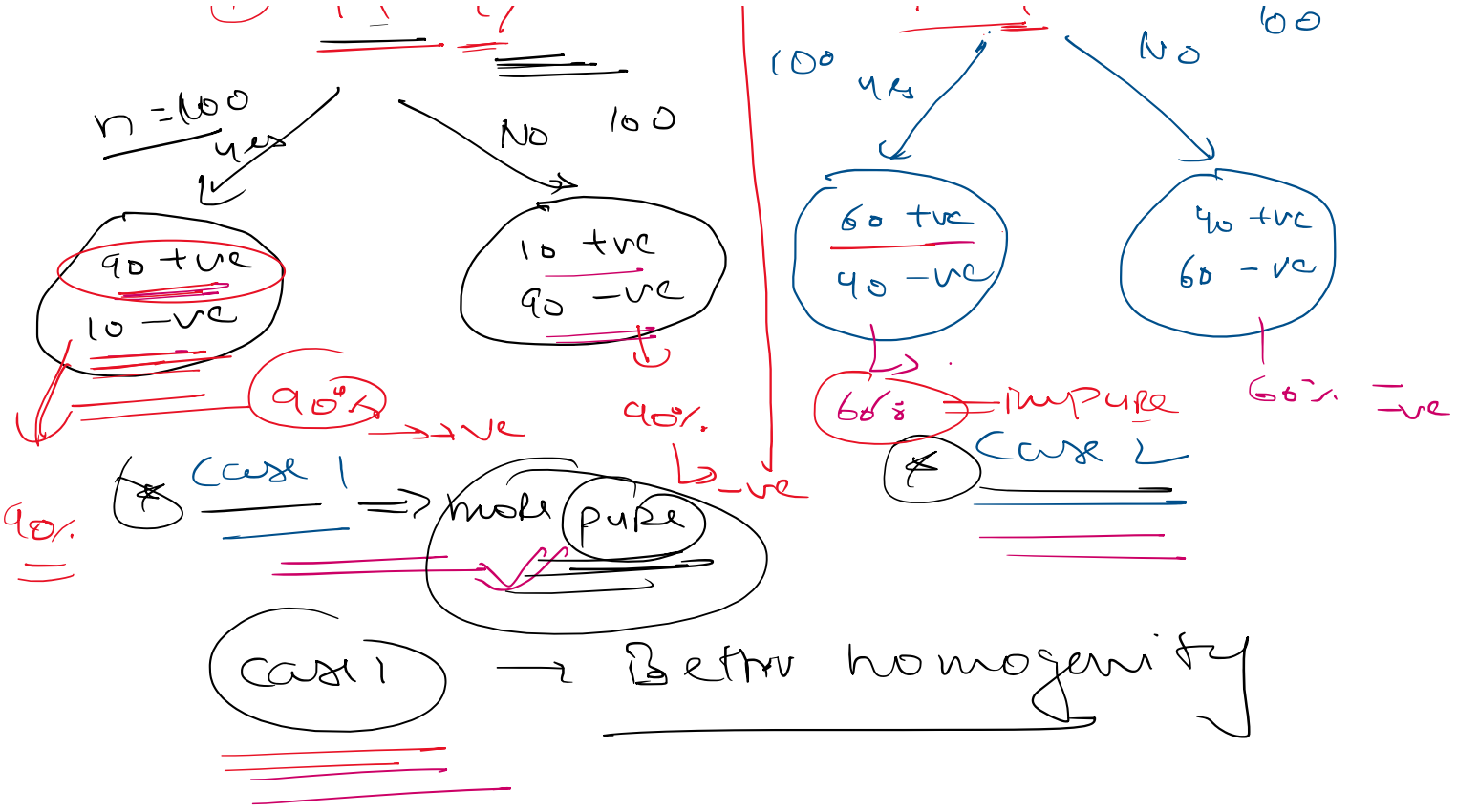
A	All statements are true	12%

A	All statements are true	12%
✓ B	II & IV	77%
C	I & III	5%
D	III & IV	7%

End Quiz Now







Case 1 → Better homogeneity

* More Homogeneity = more Better pred^n

* 10:13 ⇒

* Concept of Purity

* ENTROPY ⇒

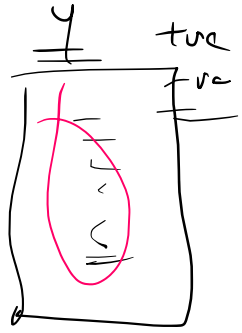
↳ measures impurity

Randomness

↳ measures (impurity)

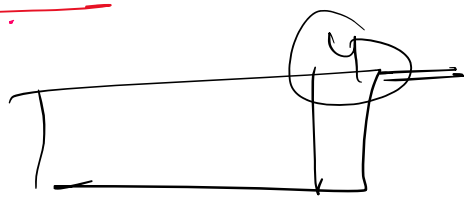
Purity = 1 − impurity

* Homogeneity



$$H(Y) = -\sum_{i=1}^{k} P(y_i) \cdot \log_2 P(y_i)$$

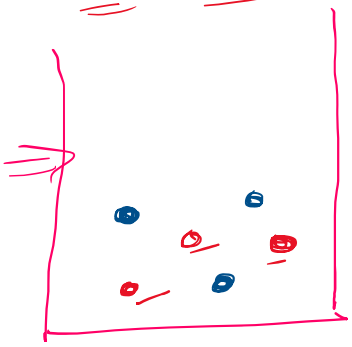
(k → no. of classes)



2

ENTROPY

3B, 3R



S_1

$$H(S_1) = -\left[P(\text{Red}) \cdot \log_2 P(\text{Red}) + P(\text{Blue}) \cdot \log_2 P(\text{Blue}) \right]$$

$$= -\left[\frac{1}{2} \cdot \log_2 \frac{1}{2} + \frac{1}{2} \cdot \log_2 \frac{1}{2} \right]$$

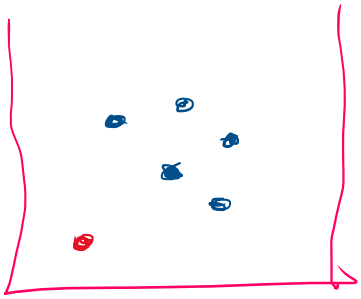
$$= -\left[\frac{1}{2}\cdot\log_2\frac{1}{2} + \frac{1}{2}\cdot\log_2\frac{1}{2}\right]$$

$$\Rightarrow +1$$

$\log_2$

impure

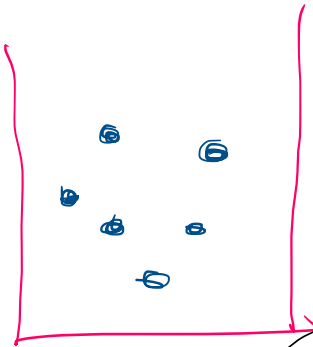
np.log



S_2

$$H(S_2) = -\left[\frac{1}{6}\times\log\frac{1}{6} + \frac{5}{6}\times\log\frac{5}{6}\right]$$

$$= 0.65$$



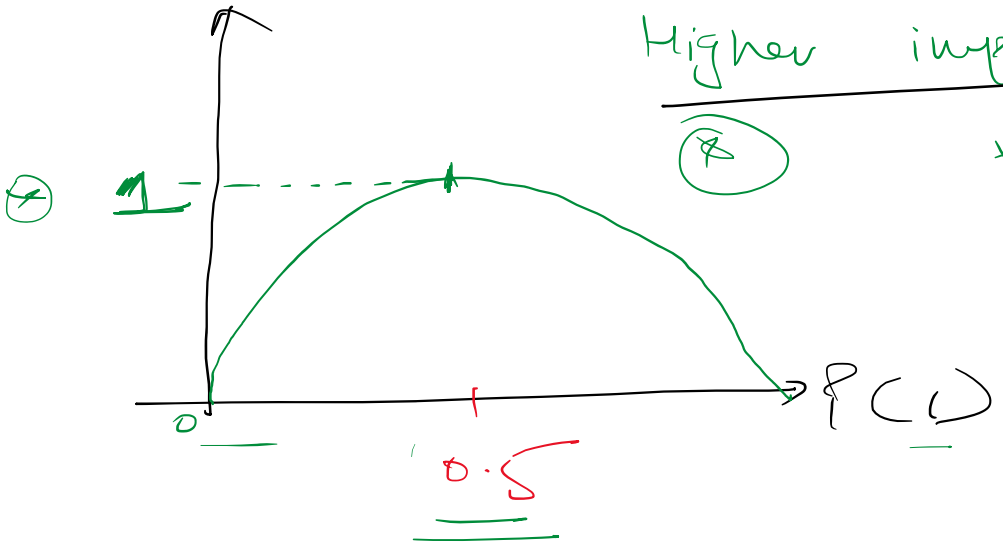
PURE

$$H(Y) = -\left(\frac{0}{6}\times\log\frac{0}{6} + \frac{6}{6}\log\frac{6}{6}\right)$$

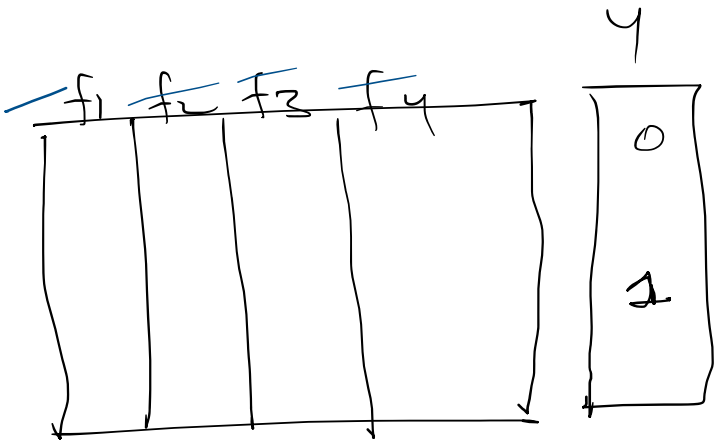
$$= -(1\log 1)$$

$$= 0$$

H = ENTROPY



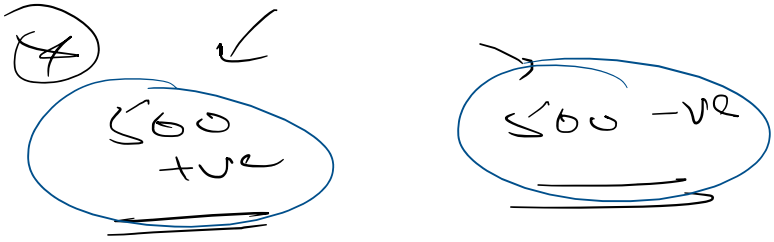
* Use this concept to create DTree



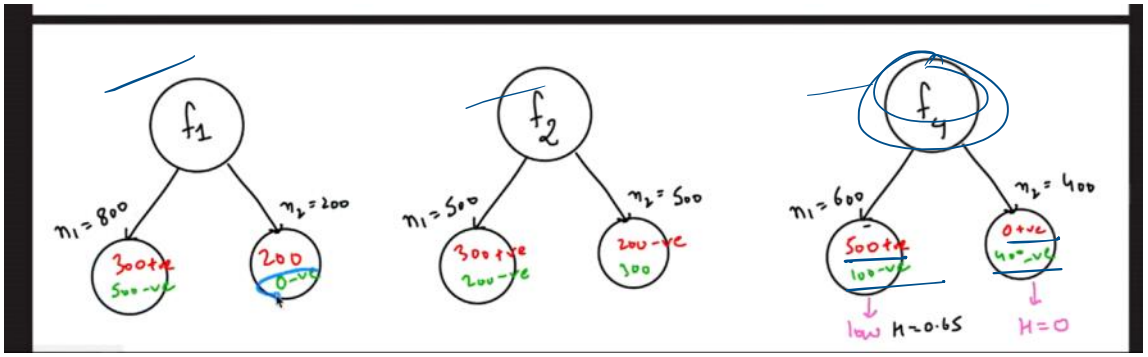
Root node

H = Entropy = 1

n = 1000 DP

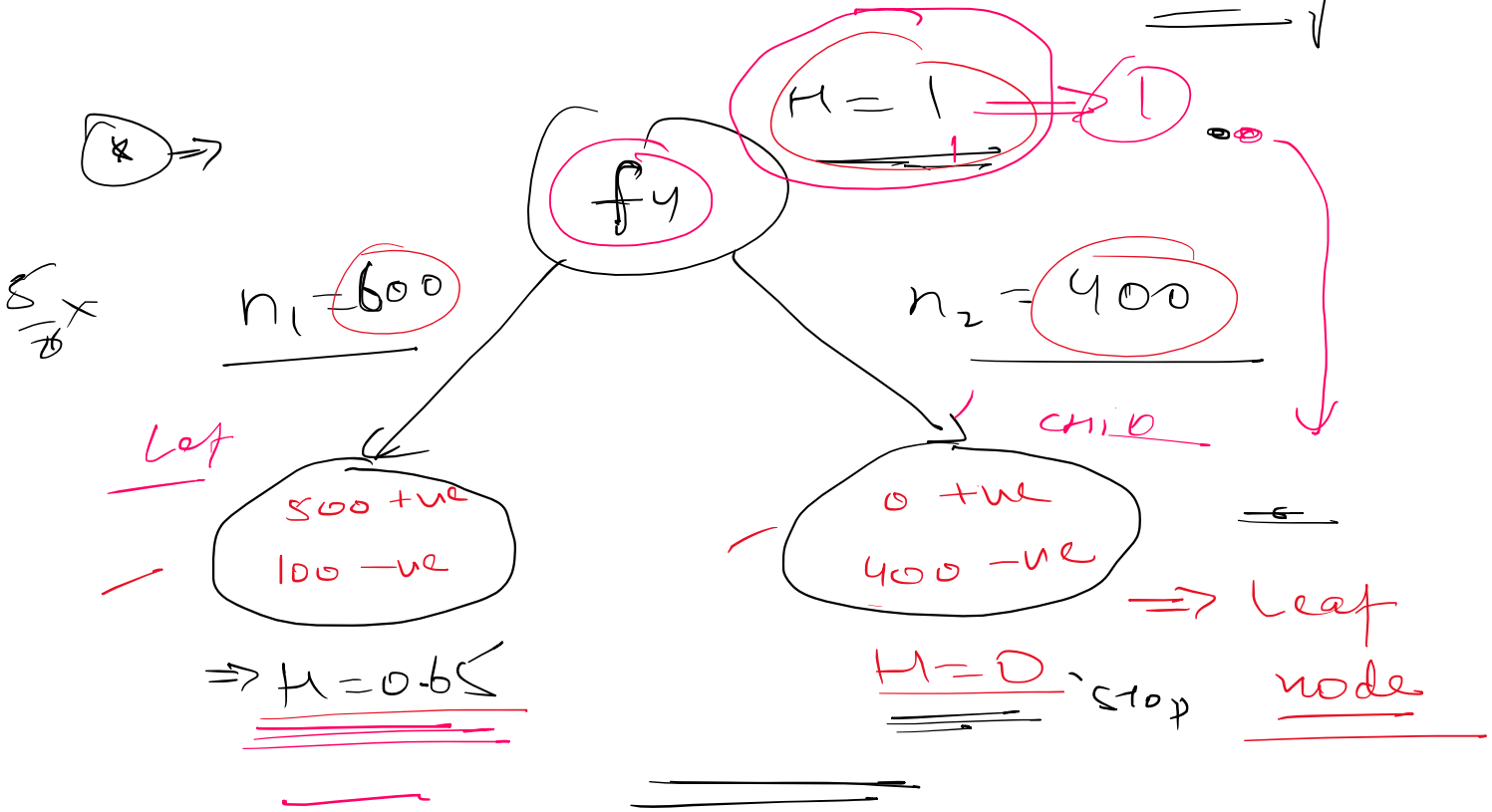


x Happy



f4

Best feature ⇒ Reduce max impurity



Ex

Entropy Reduced

f4

* Reduction in ENTROPY

for f4 ⇒ H(Parent) − H(child)

$$\Rightarrow 1 - \left[\frac{600}{1000} \times 0.65 + \frac{400}{1000} \times 0 \right]$$

$$1 - \left[0.6 \times 0.65 \right]$$

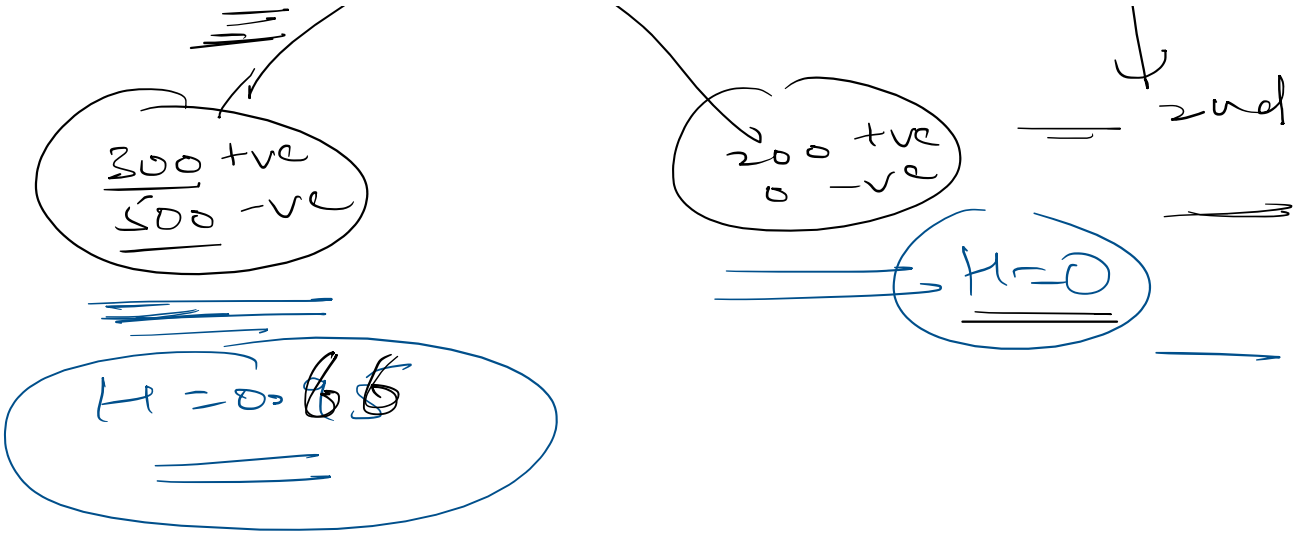
$$= 0.61$$

into gain

f1 f2

f1 ⇒ $-\left[\frac{3}{8} \log \frac{3}{8} + \frac{5}{8} \times \log \frac{5}{8} \right]$





Reduction in ENTROPY

H(Parent) − H(Child)

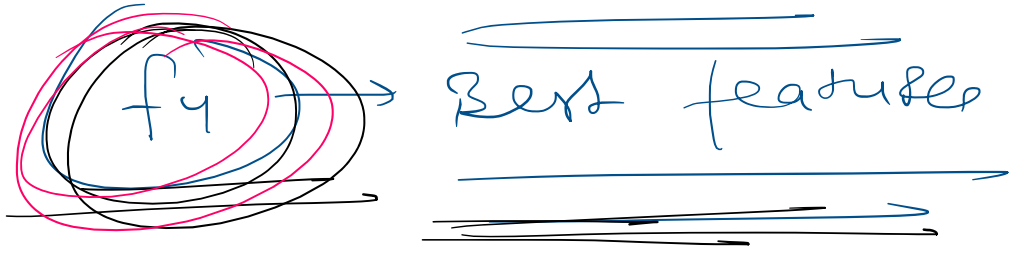
$1 - \left[\frac{800}{1000} \times 0.65 + \frac{200}{1000} \times 0\right]$

$= 1 - [0.8 \times 0.95$

$= 0.48$

$1 \longrightarrow$ $= 0.52$

Info gain

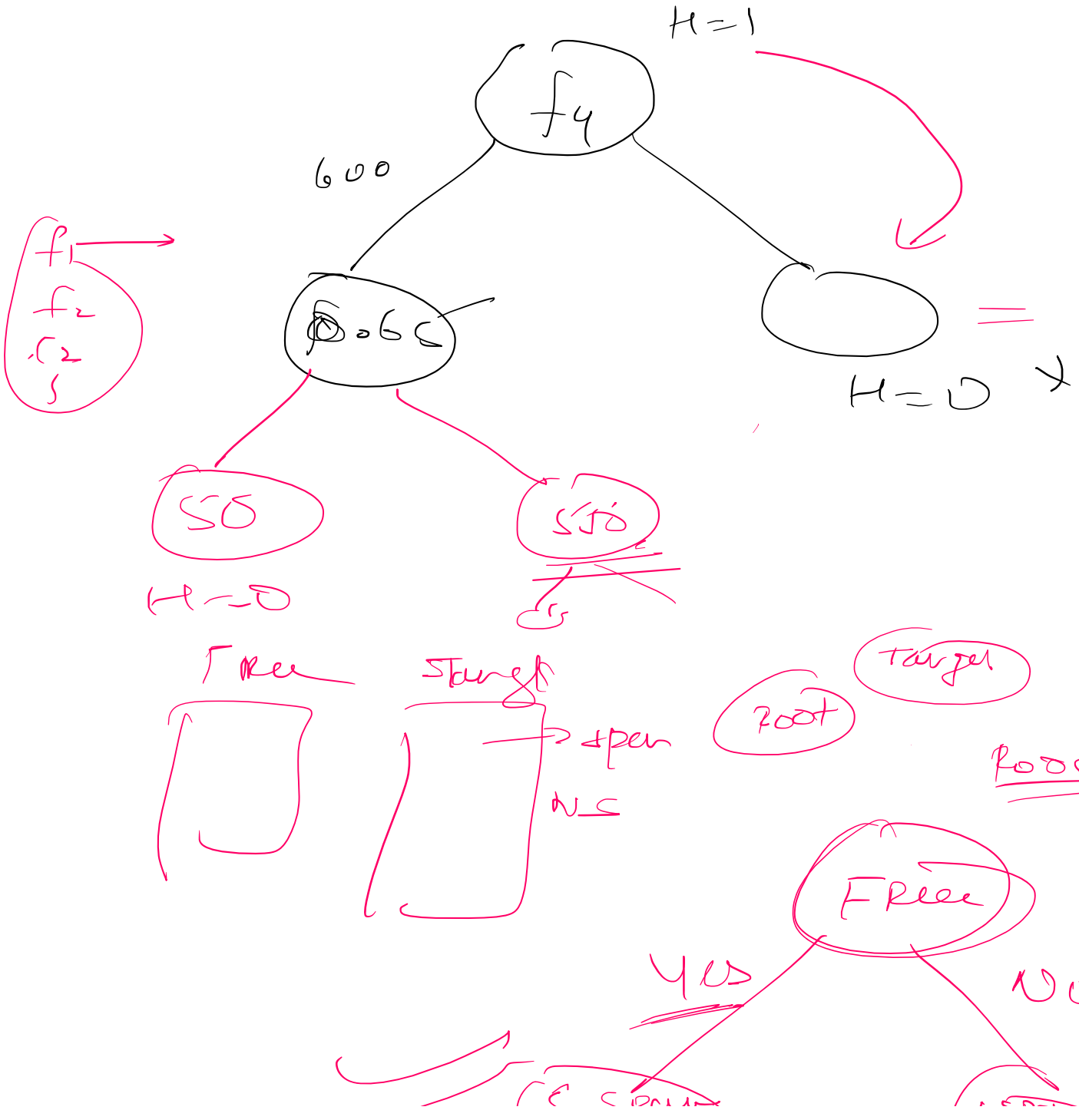


Reduc^n in ENTROPY

feature with max info gain

entropy

to be considered for split







Quiz time!

Time Left: 21s

Quiz3 - Check your understanding Which of the following statement is false ?

29 users have participated

A	Purer the node, more confidence we are in our prediction	21%
B	For making prediction, DT takes majority vote of class	17%
C	More homogenous the data is at the node, more confident we are about our prediction	45%
✓ D	None of the above	17%

End Quiz Now

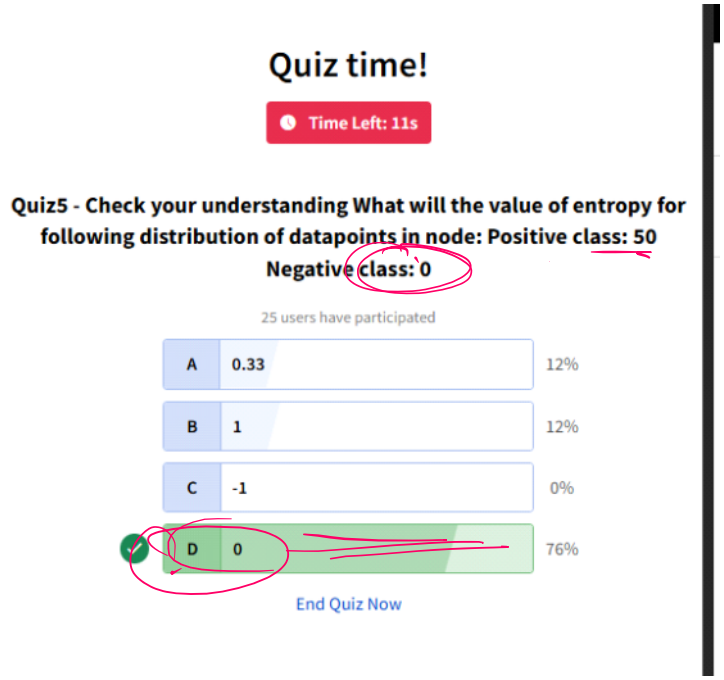


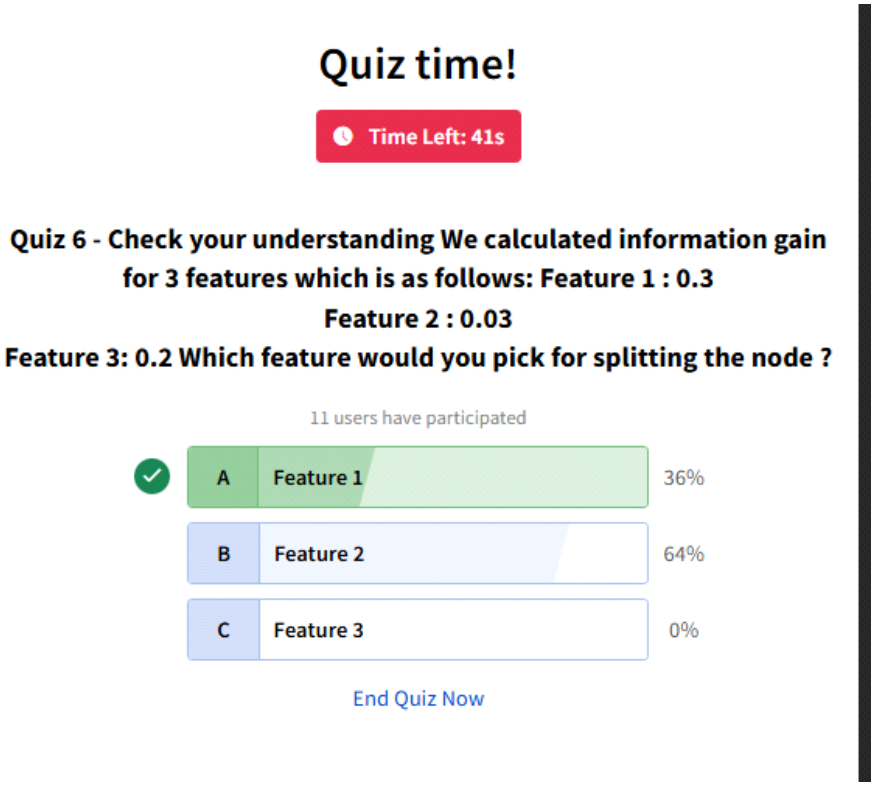
Quiz4 - Try it yourself In which of the cases will the entropy be minimum ?

12 users have participated

✓ A	A node will all datapoints belonging to one class only	58%
B	Node with datapoints belonging to both class	17%
C	Node with 100 datapoints with one datapoint belonging to positive and rest belonging to other class	25%
D	Entropy is independent of proportion of datapoints in the node.	0%

End Quiz Now


Quiz time!
Time Left: 11s
Quiz5 - Check your understanding What will the value of entropy for following distribution of datapoints in node: Positive class: 50 Negative class: 0
25 users have participated
A 0.33 12%
B 1 12%
C -1 0%
D 0 76%
End Quiz Now


Quiz time!
Time Left: 41s
Quiz 6 - Check your understanding We calculated information gain for 3 features which is as follows: Feature 1 : 0.3 Feature 2 : 0.03 Feature 3: 0.2 Which feature would you pick for splitting the node ?
11 users have participated
A Feature 1 36%
B Feature 2 64%
C Feature 3 0%
End Quiz Now